INNOVATION IN SOLID WASTE MANAGEMENT

IDLEBIN

Zero-Electricity Foot-Compacting Community Dustbin

EXECUTIVE MISSION:

"A zero-electricity, foot-operated dustbin that crushes plastic waste to stop street overflow and animal scattering in everyday Indian localities."

TEAM & INSTITUTION CREDENTIALS

Track: Waste Reduction, Segregation & Circular Economy

Team Structure: 3 Student Engineers + 1 Faculty Mentor

Target Deployment: Gram Panchayats, Tier-2/3 Municipalities & University Campuses

Grand Finale: 28 October 2026

Pondicherry University, India

The Reality: Why Indian Street Bins Fail

1. THE 80% EMPTY AIR TRAP

- Street dustbins fill up within 2 hours of placement.
- Over 80% of volume is trapped air inside uncrushed PET water bottles, paper cups, and snack packets.
- A standard 60L bin overflows after just 25 bottles.
- Trucks end up hauling empty trapped air across cities.

2. THE '10-FOOT TOSS' DISGUST BARRIER

- Traditional community bins are rusted, filthy, and foul.
- Residents refuse to touch lids with their bare hands.
- Garbage bags are thrown from 5-10 feet away.
- Bags burst on the road curb, spilling wet kitchen waste.
- Permanent 'Kachra Points' (GVPs) form on roadsides.

3. STRAY ANIMAL SCATTERING

- Once a bin is open or overflowing, cows and street dogs drag plastic bags out onto the road to reach food scraps.
- Diapers, rotting food, and plastic scatter across 50m.
- Severe disease vectors (Dengue, Cholera, Flies).
- Plastic washes directly into open gutters (naalis).

4. WHY EXISTING 'TECH BINS' FAIL

- Solar-powered smart bins cost over Rs. 3,50,000 each.
- Ultrasonic sensors get caked in mud, grime, and grease.
- Motors and lithium batteries burn out in Indian heat.
- Zero spare parts or budget in rural Gram Panchayats.
- Result: Tech bins are abandoned within 3 months.

IDLEBIN: Low-Cost Mechanical Ingenuity

CORE INNOVATION HYPOTHESIS

Do not use expensive electronics to treat the symptom. Use simple mechanical leverage to eliminate the root cause: crush empty volume instantly with zero electricity.

4X FOOT COMPACTION

- Class-1 lever mechanics.
- Downward human foot step generates 600N of force.
- Crushes PET bottles & food wrappers down by 75%.
- **Bin holds 100+ bottles instead of 25 bottles.**
- Zero street overflow.

100% HANDS-FREE

- Pure foot pedal operation.
- Zero hand contact with lid.
- Eliminates the disgust factor that causes tossing.
- Residents walk up directly and drop waste inside.
- Preserves civic hygiene and prevents bag bursts.

ANIMAL-LOCK SHIELD

- One-way gravity baffle.
- Crushing plate acts as a physical security ceiling.
- Cows and dogs cannot reach inside or pull bags.
- **Waste stays locked safely inside the drum.**
- Zero road scattering.

Technical Mechanics & Working Principle

MECHANICAL ANATOMY

1. Foot Pedal Assembly

Reinforced steel/PVC foot lever at ground level, ang
led at 25 degrees for optimal biomechanical leverage.

2. Vertical Pushrod Linkage

Solid 1/2-inch vertical transmission rod transferrin
g ground motion directly to the overhead carriage.

3. Compaction Platen (Plate)

Perforated 8mm high-density plate matching internal
drum diameter. Perforations prevent vacuum resistance.

4. Dual Tension Spring Reset

Industrial helical tension springs automatically ret
urn the platen to top resting position when foot is lifted.

5. Foul Odor Charcoal Filter

Sealed natural activated coconut charcoal tray under
lid absorbs ammonia and sulfur odors without power.

PHYSICS & SPECIFICATIONS

Mechanical Advantage (MA)	1 : 1.8 Force Amplification
Input Human Force	~350 N (Average 40 kg foot press)
Effective Crushing Force	~630 N downward platen force
Plastic Compaction Ratio	3.5:1 to 4:1 Volume Reduction
Cycle Stroke Time	< 1.5 seconds per compaction
Electrical Consumption	0.00 Watts (100% Mechanical)
Operating Temperature	-10 C to +60 C (Weatherproof)
Lifespan / MTBF	> 5 Years (Hinges & Spring replace only)

Unit Economics & Municipal Feasibility

METRIC	IMPORTED SMART BIN	OPEN MUNICIPAL DUMPSTER	IDLEBIN (OUR INNOVATION)
Unit Capital Cost (CapEx)	Rs. 3,50,000+	Rs. 3,000 - 4,500	**Rs. 650 - Rs. 850 (Ultra-Low)**
Power Source	Solar Panel + Lithium Batt.	None	**None (Zero Electricity)**
Effective Capacity	Compacted (Motorized)	2-3 Hours before overflow	**4X Compacted Capacity**
Maintenance Difficulty	Requires OEM engineers	Low (Rusts & dents)	**Zero (Local blacksmith/carpenter)**
Animal Attack Resistance	Moderate	Zero (Cows/dogs scatter)	**100% Animal-Proof Baffle**
Rural Feasibility	0% (Prohibitive Cost)	Poor (Creates Black Spots)	**100% Feasible for Panchayats**

MUNICIPAL BUDGET EFFICIENCY: For the price of ONE imported smart solar bin (Rs. 3.5 Lakhs),
a municipality or Gram Panchayat can deploy 450+ IDLEBINS, covering an entire town.

Ground Impact & Swachh Bharat Alignment

1. ELIMINATION OF GVPs (BLACK SPOTS)

Under Swachh Bharat Mission (SBM-U 2.0), eliminating Garbage Vulnerable Points is a priority KPI. IdleBin prevents the overflow that sparks roadside dump creation, keeping roads permanently clean.

2. WORKER DIGNITY & ERGONOMICS

Sanitation workers no longer have to shovel rotten, scattered garbage mixed with dirt using hand s hovels. Compacted trash is lifted out in dense, clean bags, reducing occupational injuries and dust inhalation.

3. 50% FUEL & LOGISTICS REDUCTION

Because waste density increases 4x, municipal auto-tippers and tractors need 50% fewer daily colle ction runs per ward, directly saving thousands of liters of municipal diesel and cutting carbon emissions.

4. TRUE PAN-INDIA SCALABILITY

From high-density pilgrimage centers in Varanasi and Madurai to quiet agricultural villages in Tam il Nadu, IdleBin operates identically without requiring mobile signals, grid power, or technical training.

Grand Finale Demonstration (Pondicherry University)

LIVE TABLETOP DEMO ON 28 OCT

Step 1: The Touchless Foot Press

Demonstrating effortless foot pedal actuation on stage with zero hand contact.

Step 2: Live Crushing Test

Placing 3 uncrushed PET water bottles into the bin. Stepping on the pedal once w

Step 3: Proof of 75% Compaction

Opening the inspection hatch and showing the flattened bottles to the jury panel.

Step 4: Animal Security Test

Demonstrating how the gravity baffle prevents reaching in from the outside.

WHY IDLEBIN IS UNBEATABLE

100% Demonstration Reliability

Unlike IoT projects that fail due to venue Wi-Fi, dead batteries, or projector lag, pure mechanics never fails on stage.

Authentic Ground Truth

Addresses the exact, ugly street problem that every ju dge encounters on their daily commute to the university.

Immediate Pilot Viability

Can be piloted on the Pondicherry University campus ca nteen within 48 hours to crush juice cups and bottles.

Zero Fabrication Barrier

Built using common workshop tools, accessible to any e ngineering team across India.